

Raagav Ramakrishnan

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EDUCATION

International Institute of Information Technology, Hyderabad (IIIT-H) 2022 – 2026
B.Tech Honours in Electronics and Communication

The Millennium School, Dubai June 2022
High School Diploma CBSE — Grade 10: 93%, Grade 12: 95%

PROJECTS

Vision Transformers | *PyTorch* Apr. 2025

- Implemented from-scratch Vision Transformer (ViT) and Differential ViT on CIFAR-10.
- Benchmarked 11 ViT configurations (patch sizes 2, 4, 8; augmentation strategies), achieving **83.69%** test accuracy.
- Extended to DiffViT with differential attention, attaining **84.38%** test accuracy.
- Visualized attention maps (DINO, rollout) and analyzed hyperparameter impacts.

Fully Convolutional Networks for Semantic Segmentation | *PyTorch, VGG* Apr. 2025

- Implemented FCN-32s, 16s, 8s architecture with skip connections on a 13-class street scene dataset.
- Fine-tuned a VGG16 backbone with bilinear upsampling, with a test mIoU of 83.3% on FCN-8s.
- Developed modular PyTorch training pipeline with per-class IoU metrics and real-time learning curve visualizations to evaluate semantic segmentation performance across model variants.
- Demonstrated impact of skip connections and backbone fine-tuning across low-frequency and small-object classes (e.g., Pole, Traffic Sign).

CNN-RNN OCR | *Python, NumPy, PyTorch* Oct. 2024

- Developed a hybrid Deep Learning pipeline for OCR tasks using CNNs for feature extraction and RNNs with integrated attention mechanisms.
- Constructed a custom dataset of word-image pairs with normalization, and label encoding.
- Achieved 99.1% ANCC (character-level accuracy) on synthetic OCR.

Multi-Layer Perceptron (MLP) from Scratch | *Python, NumPy, Matplotlib* Sept. 2024

- Implemented a fully customizable MLP from scratch using Python and NumPy, to better understand the fundamental operations behind neural networks.
- Supported flexible configurations, including customizable hidden layers, activation functions (ReLU, Sigmoid, Tanh), and optimization methods (SGD, Mini-batch and Batch Gradient Descent).
- Integrated loss functions like MSE and Cross-Entropy Loss for tasks like regression and classification.
- Gained deeper insight into backpropagation and gradient descents in the neural network architecture by manual implementation.

BPSK Modulation System | *MATLAB* April 2024

- Built a Binary Phase Shift Keying (BPSK) system for audio transmission over an AWGN and Noiseless channel, using rectangular and raised cosine pulse shaping modulation schemes for ISI analysis.
- Analyzed system performance through BER metrics and constellation diagrams.

Y86-64 ISA Processor | *Verilog* Feb. 2024

- Designed and simulated a Y86-64 instruction-level processor with control flow and memory operations.
- Built a pipelined implementation with intermediate registers to optimize instruction throughput along with hazard detection and solutions to decode the same.
- Validated functional correctness through waveform-based testbenches and timing verification.

- Implemented a 4-bit ALU starting with Verilog modules, progressing to gate-level simulations in NGSpice, and finally transistor-level designs (NMOS/PMOS) and layout verification using MAGIC.

WORK EXPERIENCE

Undergrad Researcher

Jan. 2025 – Present

Data Sciences and Analytics Center (DSAC), IIIT Hyderabad

Advisor: Dr. Nimmi Rangaswamy

- Conduct an interdisciplinary honors research project titled “The Role of Technology in Shaping Perceptions of Well-Being in Fitness and Body-Building”.
- Investigate how fitness tech and social media influences workout routines, body image, and well-being, along with challenges in personalization, cultural relevance, and user engagement to guide inclusive, context-aware wellness tech
- Leading a comprehensive literature review and conducting ethnographic interviews with individuals aged 18–30 to gather primary qualitative data.

ML Research Engineer

Dec. 2024 – July 2025

Cloud Ambassadors (Startup within the Google Ecosystem)

- Fine-tuning Protein Transformers (ProtBert and ESM-1b) to predict mutations in protein sequences.
- Explored Protein folding mechanism and mutation prediction, Homology, AlphaFold v2, and utilizing Graph based DL Architecture (Graphnormers and Message Passing Neural Networks).
- Developed a live video dubbing pipeline with Google, OpenAI and ElevenLabs solutions for real-time multilingual speech translation.
- Designed regex-based parsing modules for language-specific text normalization and phonetic processing for Gujarati and Hindi.
- Built an end-to-end Dynamic Pricing Engine using Kubeflow pipelines, BigQuery, and MongoDB, integrating XGBoost models to predict optimal flight prices.
- Automated training and inference workflows via Vertex AI and Kubeflow, with Pub/Sub and Cloud Build triggers for seamless model deployment.

ADDITIONAL EXPERIENCE

Debate Club, IIIT-H - President

Oct. 2023 - Present

- Host Debate and MUN tournaments with 50+ attendees; manage logistics, marketing, and adjudication, while building specialized teams for the same.
- Organized bi-weekly meetings, mock debates, and training modules to foster a competitive debate culture on campus, and it's grown steadily since the club has been revamped.

TEDx Talk: “Clarity in a Changing World”

March 2022

ML Certificate (Udemy)

June 2024

Design Thinking (IDEOU)

August 2021

- Completed IDEOU Foundations in Design Thinking with applied user research projects.

SKILLS

- **Languages:** C, C++, Python, R, JavaScript, HTML, CSS, Bash, L^AT_EX
- **Hardware:** Verilog, SPICE, MAGIC Circuit Layout, Oscilloscopes, Arduino
- **ML/DS Tools:** PyTorch, Transformers, VertexAI, Sci-kit Learn, NumPy, Pandas
- **Systems:** Git, Ollama, Docker, MongoDB, Google Cloud Platform, Kubeflow
- **Soft Skills:** Public speaking, Technical writing, Team leadership, Cross-functional communication
- **Spoken Languages:** English, Tamil, Hindi, Urdu